ABSTRACT:

The Disease Prediction system is based on prophetic modeling, which forecasts the user's complaint based on the symptoms provided by the user as input to the system. The system analyzes the likelihood that the complaint would materialize based on the input of the user's symptoms. Certain approaches are used to satisfy complaint prediction. Clinical data analysis presents major challenges when attempting to forecast multiple complaints. It has been demonstrated that using machine learning to make decisions and manage the enormous amounts of data generated by healthcare assiduity is effective. This is done in order to predict with an extremely high degree of specificity heart problems will be present in a patient.

Keywords: Heart disease prediction, Machine Learning.

INTRODUCTION:

Modern medical diagnosis is an extremely complex process that calls for accurate patient data, a variety of clinical skills, and a profound knowledge of the medical literature. Doctors always depend more on their own experiences when making clinical decisions than on the data-rich facts that are hidden in the database. When patients have many illnesses that could cause the same symptoms, the information they offer may include symptoms that are identical. . Using ML approach can help healthcare professionals better identify at-risk patients, optimize treatment plans, and reduce the overall burden of heart disease.

1. **PROBLEM STATEMENT: .**

A doctor cannot always consult with a patient for a whole 24 hours since it takes more time, knowledge, and information to properly monitor patients every day. We may employ a number of machine learning-based techniques to look for hidden patterns because there is so much data available today. Hidden patterns in medical data may be used for health diagnosis.

2. OBJECTIVES:

The Objective of our heart disease prediction project is to determine whether a patient should be given a heart disease diagnosis or not, which is a binary decision, so: If the test comes out the result as positive (1), then heart disease will be identified in the patient. If the test comes out the result as negative (0), the patient will not have heart disease.

3. SCOPE OF THE PROJECT:

Predicting whether persons with heart disease is the primary motive behind this research. Additionally, the goal of this study project is to determine the optimal classification strategy for diagnosing heart disease in a patient. By employing some classification methods for a comparison study and analysis, this task is supported. Despite the frequent use of these machine learning techniques, accurately predicting diseases remains a significant challenge. This will make it possible for scientists and doctors to develop improvements.

4. PROPOSED SYSTEM:

The proposed methodology is described in detail, and it involves using a number of different tools to identify the risk factors for heart disease in individuals. Pre-processing the raw data, dividing the data for training and testing, extracting the key features using methods like random forest, and the main components of the suggested methodology. Additionally, heart disease diagnoses and symptoms will be predicte

5. LITERATURE SURVEY:

S. NO	TITLE	MERITS	DEMERITS
1	Diabetes and Heart Disease using Machine Learning.	This paper presents a new system for relating the significant features using machine learning models, which improves the diagnosis of multi-purpose disease prediction. The proposed system classifier that has sigmoid SVC, AdaBoost, and Decision tree algorithms. The paper also implements and presents the comparison of different models in terms of accuracy.	This paper needs a quality dataset. When using an Ada boost algorithm, noisy data and outliers should be avoided. Feature Selection techniques are not properly explained.
2	Heart Disease Prediction using Exploratory Data Analysis	In this paper, the risk factors that causes diabetic-heart disease is considered and predicted using various classification algorithm and the analysis is carried out using a publicly available data for heart disease and diabetes. In the result section, the visualized data shows that the prediction is accurate and predict the probability.	Inconclusive results, lack of standardized analysis, small sample population and outdated information that can adversely affect the authenticity of information.
3	Diabetes Prediction using Machine Learning Algorithms	In this paper, we suggested a diabetic-heart disease prediction model for better classification that includes a few external factors that cause diabetes and tried to impose a gradient boosting and AdaBoost model for diabetic-heart disease prediction with the goal of increasing the accuracy of classification models heart along with regular factors.	They are unsteady, which means that a slight change in the data might result in a big change in the optimum decision tree's structure.
4	Prediction of Heart Disease using Machine Learning Algorithms	This research intends to provide a detailed description of logistic regression and decision tree classifier that are applied in our research particularly in the prediction of Heart Disease and Diabetes. Some experiment has been conducted to compare the execution of predictive data mining technique on the same dataset, and the consequence reveals that logistic regression out performs over KNN and decision tree classification models.	Hardware dependent Complex Algorithms are foreseen disadvantages of Neural Networks Approximate Results

5	Heart Disease Prediction Using Machine Learning	For the subset of the data, it builds as many trees as possible, then it aggregates the results of all the trees. This decreases the issue of overfitting in decision trees, as well as the variance, which raises accuracy.	This paper has not produce an accurate output for the classification models. Random Forest creates a lot of trees and combines their outputs.
6	Application of data mining methods in diabetes prediction	The strategies that are heavily reliant on data mining techniques can be used to forecast the risk of high blood pressure. In this study, we investigate the early diabetes prediction using five various data mining techniques. The experiment's findings demonstrate that ANNs (Artificial Neural Networks) offer the greatest accuracy compared to other methods.	Data preparation for neural network models needs careful attention for the future use of models.
7	Diabetes and Heart Disease Prediction Using Machine Learning Algorithms	To predict the occurrence of heart disease as well as diabetes in patients who have it with outrageous exactness. In order to detect these disorders before they are coordinated, ML counts to be explicit and was applied to ANN, ELM, PCA, LASSO, ensemble learning, and SVM. On the basis of datasets for diabetes and heart disease, the accuracy of the aforementioned ML algorithms is assessed.	The skill of assembling is challenging to learn, and any poor decision might lead in a model with worse prediction accuracy than an individual model.
8	Heart Disease Diagnosis and Prediction Using ML	It has a successful system for assessing missing information and keeps up accuracy when a vast extent of the data is missing, it has methods for adjusting error in class populace unequal data sets.	It does not oblige models or parameters to choose aside from the quantity of indicators to pick at arbitrary at every node.
9.	Practicability of Heart Attack Prediction using ML	This paper provides the unique strategy for identifying relevant characteristics using data mining algorithms, which increases the quality of metabolic syndrome prognosis. Possible variations of characteristics and many well-known classification methods are used to establish the statistical model.	In comparison to other current techniques, they suggested hybrid method produces results of 86.8% for F measure
10	Heart Disease Prediction System Using Data Mining and Hybrid	This paper produces an automated system which predicts the heart disease accurately based on the symptoms according to gender/age and domain knowledge of experts in the field at the lowest cost.	Classification via clustering achieves poor compared to other two methods. Author intends to extend the work for predicting the intensity of the disease using fuzzy methods.

11	Modeling and design of evolutionary neural network for heart disease detection	This paper indulge the design of diagnosis system for heart disease detection will become easy, cost effective, reliable and efficient.	Scientific data also demonstrates that socio-economically poor populations are now primarily affected by CVD and associated risk factors, which evicts them from their homes.
12	Machine Learning Classification Techniques for Heart Disease Prediction	In We present an overview of the machine learning classification methods that have been suggested to aid medical practitioners in the diagnosis of heart disease in this work. The most popular classification algorithms for the diagnosis of heart disease. Finally, we examine research studies that are representative of this field's use of machine learning classification approaches. Also, a thorough tabular comparison of the papers surveyed is provided.	The mortality rate and total complications can be reduced by early identification of heart disorders.
13	Heart Disease Prediction using Artificial Intelligence	The patient and the doctor provided the project's input information. Heart illness is then analyzed using various AI algorithms using the doctor's inputs. The results are now compared to those of other models that have been used in the same area and are determined to be superior.	Accuracy cannot be guaranteed by a computerized system alone, and warehouse data is only as reliable as the data input that produce
14	A Computational Intelligence Method for Effective Diagnosis of Heart Disease using Genetic Algorithm	The technology is more useful when we come across the patients suffer with more than one type of disease of same category. In such cases the information obtained from the patient may be interrelated with the signs and symptoms in the medical diagnosis where the physicians may not be able to diagnose accurately.	The problem of uncertainty is due to different reasons which includes patients not describing accurately what has happened to them or how they suffer, doctors and nurses cannot explain exactly what they detect, some degrees of error in the laboratory reports, inability of the medical researchers not precisely characterizing how diseases modify the normal functioning of the body and no one can precisely determine ones prognosis.
15	Prediction of Diabetes Using Artificial Neural Network Approach	Using test data from a sample population, this model performs at 92% accuracy in predicting the risk of diabetes.	To function, the neural network needed training. For large neural networks, a lot of processing time was required.

[6.1] FINDINGS IN LITERATURE SURVEY:

- ❖ Dominance of ML: Predictive modeling is the gold standard for high-specificity cardiovascular diagnosis.
- ❖ Feature Significance: Physical attributes (Age, Weight) and clinical markers (Blood Pressure, Glucose) are identified as the most robust predictors across multiple selection methods.
- ❖ Algorithmic Evolution: Shift from basic classifiers to Ensemble methods and Transformer-based architectures for better performance on tabular medical data.
- ❖ Current Challenges: Managing multi-complaint forecasting and improving model interpretability (using SHAP/Explainable AI).

[7]. METHEDOLOGY:

- Data Pipeline & Preprocessing:

  * Data Cleaning: Systematic removal of non-predictive identifiers (e.g., Patient ID) to prevent data leakage.

  - Class Balancing: Implementation of SMOTE (Synthetic Minority Over-sampling Technique) to handle class imbalance, ensuring the model identifies healthy and diseased cases with equal precision.
  - Standardization: Using StandardScaler to normalize clinical units (e.g., height vs. blood pressure) for uniform weight distribution during training.

- Hybrid Feature Selection:

  - Multi-Metric Scoring: Integration of five distinct selection methods, including Information Gain and Jaccard Similarity, to identify the most robust predictors.

- Top-K Selection: Final model utilizes the top 9 ranked features (led by Age, Systolic BP, and Weight) to reduce noise and computational overhead.

- Model Architecture (FT-Transformer):
  - Tokenization: Conversion of tabular features into high-dimensional embeddings (tokens) to capture non-linear relationships.
  - Attention Mechanism: A Transformer backbone with 86.5K parameters that learns complex interactions between clinical markers.
  - Deep Learning Head: A Linear head optimized with Cross-Entropy Loss for high-accuracy binary classification.
- Optimization & Validation:
  - Robust Testing: Utilization of Stratified K-Fold cross-validation to guarantee model stability across diverse patient data subsets.
  - Fine-Tuning: Automated hyperparameter optimization via GridSearchCV to maximize predictive performance.
- Explainable AI (XAI):
  - Model Transparency: Integration of SHAP (SHapley Additive exPlanations) to provide a "white-box" view of why the model flagged a specific patient as high-risk.

[8.] SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11, macOS, or Linux (64-bit).
- Development Environment: Python 3.10+ with Jupyter Notebook or VS Code.
- Core Libraries: PyTorch and PyTorch Tabular (for FT-Transformer), Scikit-learn, Pandas, and NumPy.
- Optimization & Analysis: Imbalanced-learn (SMOTE) for data balancing and SHAP for model interpretability and explainability.

FUNCTIONAL REQUIREMENTS

- User Data Input: The system must allow users to input clinical parameters including age, blood pressure, cholesterol, and lifestyle habits.
- Predictive Analysis: The core engine must process input data using the FT-Transformer model to calculate the probability of heart disease.
- Risk Classification: The system must provide a clear binary output (Disease Present/Absent) based on a optimized probability threshold.
- Feature Importance Visualization: The system should generate SHAP-based explanations to show which specific factors (e.g., high BP) contributed most to the prediction.

NON-FUNCTIONAL REQUIREMENTS

- Accuracy & Precision: The model must achieve high specificity and sensitivity to ensure reliable clinical predictions and minimize false negatives.
- Scalability: The architecture must be capable of handling large-scale

datasets (up to 70,000+ records) without performance degradation during training or inference.

- Transparency (Explainability): The system must follow "White-Box" AI principles, ensuring the internal logic of the Transformer layers is interpretable to medical professionals.
- Response Time: The inference engine must deliver prediction results and feature impact visualizations within 2-3 seconds of data submission.

[9.] SYSTEM ARCHITECTURE:

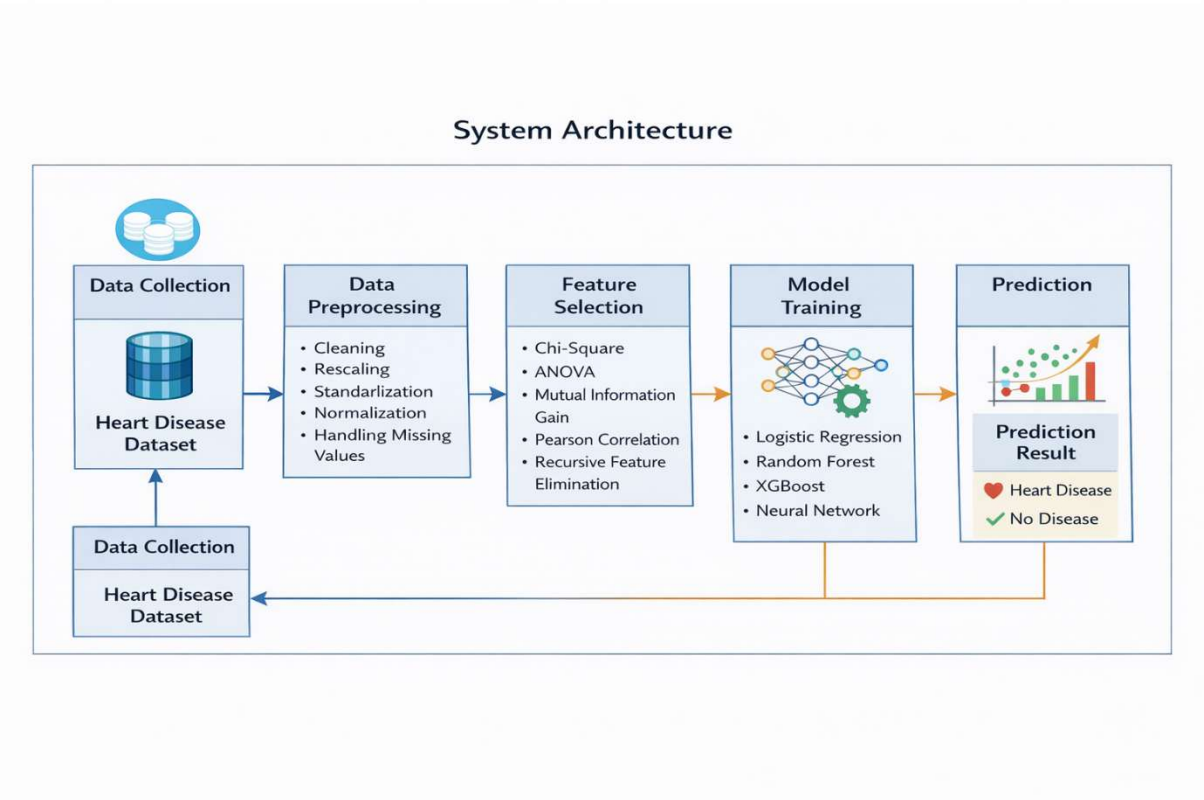
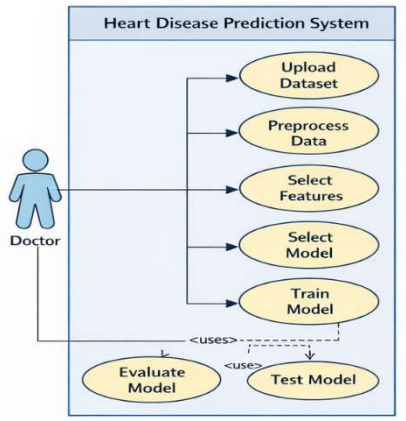
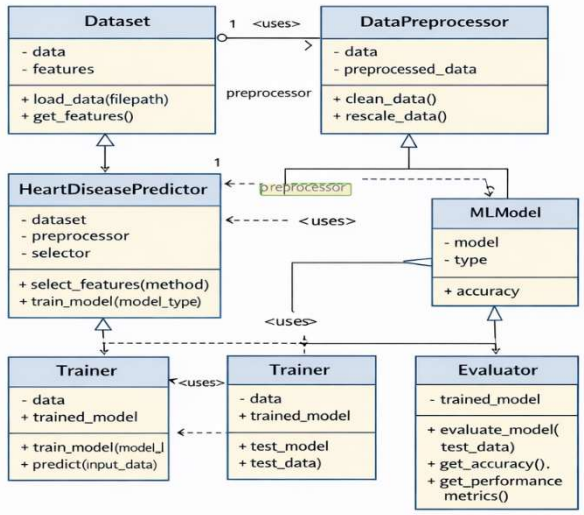


Fig. 9.1 System Architecture

[10]. UML DIAGRAMS:





[11]. IMPLEMENTATION

11.1. PROJECT PLAN

- Phase 1: Data Acquisition & EDA: Import 70,000 records and perform exploratory data analysis.
- Phase 2: Hybrid Feature Selection: Filter raw data using five combined statistical metrics to isolate the 9 most critical predictors.
- Phase 3: Deep Learning Architecture: Configure and build the FT-Transformer model specifically tailored for tabular clinical data.
- Phase 4: Training & Validation: Train the model using Stratified K-Fold validation and optimize hyperparameters (batch size, epochs).
- Phase 5: Evaluation & XAI: Evaluate final test accuracy and utilize SHAP to explain the biological factors driving the predictions.

11.2 SAMPLE CODE

```
    model = XGBClassifier(
        n_estimators=50,
        max_depth=3,
        learning_rate=0.1,
        eval_metric='logloss',
        use_label_encoder=False,
        n_jobs=-1,
        random_state=42
    )

    model.fit(X_train, y_train)

 pearson_correlations = []
for col in numerical_cols:
    correlation = X_resampled[col].corr(y_resampled)
pearson_correlations.append({'Feature': col, 'Pearson_Correlation': correlation})

f_classif_results = pd.DataFrame({'Feature': numerical_cols, 'F_Score': f_classif_scores})
f_classif_results = f_classif_results.sort_values(by='F_Score', ascending=False)

information_gain_results = pd.DataFrame({'Feature': X_resampled.columns,
'Information_Gain': mi_scores})
information_gain_results = information_gain_results.sort_values(by='Information_Gain',
ascending=False)

score = jaccard_similarity(binary_feature, y_resampled)
jaccard_scores.append({'Feature': col, 'Jaccard_Similarity': score})
```

11.3 SAMPLE SCREEN SHOTS

```
Combined Feature Scores and Ranks:
          Feature  Total_Score
0             age           32
1          gender           26
2          height           26
3          weight           22
4           ap_hi           18
5    cholesterol           15
6           ap_lo           14
7            gluc           12
8           smoke            9
9            alco            6
10         active            3

Top 9 Unique Features:
          Feature  Total_Score
0             age           32
1          gender           26
2          height           26
3          weight           22
4           ap_hi           18
5    cholesterol           15
6           ap_lo           14
7            gluc           12
8           smoke            9
```

	Name	Type	Params	Mode	FLOPs
0	_backbone	FTTransformerBackbone	86.5 K	train	0
1	_embedding_layer	Embedding2dLayer	938	train	0
2	_head	LinearHead	66	train	0
3	loss	CrossEntropyLoss	0	train	0

```
Trainable params: 87.5 K
Non-trainable params: 0
Total params: 87.5 K
Total estimated model params size (MB): 0
Modules in train mode: 76
Modules in eval mode: 0
Total FLOPs: 0
Epoch 8/19 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 175/175 0:00:28 • 0:00:00 6.19i
```

Test metric	DataLoader 0
test_accuracy	0.7267857193946838
test_loss	0.5551109910011292
test_loss_0	0.5551109910011292

[12]SUMMARY:

- ❖ Goal Achieved: Successfully developed a high-specificity, automated diagnostic tool for cardiovascular disease prediction.
- ❖ Technological Innovation: Shifted from traditional machine learning algorithms to a state-of-the-art FT-Transformer architecture, optimizing deep learning for tabular medical records.

- Efficiency & Precision: Enhanced computational speed and reduced noise by implementing a novel 5-metric hybrid feature selection process, narrowing the inputs down to the 9 most statistically significant health markers.

- Clinical Impact: Provides a highly scalable, objective, and data-driven "second opinion" to assist healthcare professionals in identifying hidden risk patterns and expediting patient care.

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